

Computational evidence for the optimality of Conway–Kochen’s 31-vector Kochen–Specker set

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We report a systematic computational search for Kochen–Specker (KS) sets with fewer than 31 vectors in dimension 3, where the Conway–Kochen construction (CK-31) has stood as the smallest known for over three decades and the best proven lower bound is 24 vectors. No sub-31 KS set is found by any of six complementary strategies: SAT-based minimization, cross-pool mixing, expanded coordinate alphabets, numerical optimization, exhaustive criticality testing, and triad density analysis. A key structural finding is the *norm-2 boundary*: integer alphabets lacking the coordinate ± 2 (such as $\{0, \pm 1, \pm 3\}$) are not KS-uncolorable at all, confirming that the cancellation identity $1 + 1 = 2$ is essential. We establish that CK-31 is 3-critical (all 4991 subsets of size ≥ 28 tested) and that cross-pool orthogonalities never contribute to minimal sets. Vertex merging produces 15 abstract KS-uncolorable graphs on 30 vertices, but Z3 cannot resolve their $\mathbb{R}^3$ realizability, confirming that the 24–31 gap is fundamentally a realizability problem.

INTRODUCTION

The Kochen–Specker (KS) theorem [1] demonstrates that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories. In dimension 3, the theorem requires exhibiting a finite set of rays (directions) such that no consistent $\{0, 1\}$ -valued coloring assigns exactly one 1 per orthonormal basis while respecting pairwise orthogonality exclusion. The minimum number of rays needed for such a construction is a fundamental open problem.

Conway and Kochen’s 31-vector set [2], which uses integer coordinates from the alphabet $\{0, \pm 1, \pm 2\}$, has stood as the smallest known KS set in dimension 3 for over three decades. The best lower bound in dimension 3 is 24 vectors, established by Li, Bright, and Ganesh [3] using SAT-based methods combined with algebraic realizability checking. Closing the gap between 24 and 31—or proving that 31 is optimal—remains open.

Recent work [4] identified six algebraic number rings that support KS constructions in dimension 3, each producing a distinct ray pool. This algebraic classification provides a structured framework for an exhaustive sub-31 search: rather than searching over arbitrary ray configurations, we can systematically test every known algebraic source of KS orthogonalities.

METHODS

We tested six complementary strategies, each designed to probe a different region of the search space.

Strategy 1: Exact minimization. For each of the six known algebraic ray pools (integer, Eisenstein $\mathbb{Z}[\omega]$, Peres $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[\sqrt{-2}]$, Heegner-7 $\mathbb{Z}[(1+\sqrt{-7})/2]$, golden ratio $\mathbb{Z}[\varphi]$), we applied three minimization methods: (a) MUS (minimal unsatisfiable subset) extraction from the SAT encoding, yielding a locally minimal (not nec-

essarily globally minimum) ray set; (b) enhanced greedy deletion with 2000 random trials plus five deterministic orderings (degree ascending/descending, triad participation ascending/descending, combined); (c) swap-based local search starting from greedy minima. The SAT encoding uses Glucose4 [5] via PySAT, with both triad constraints (exactly one ray per basis colored 1) and pairwise exclusion constraints for all orthogonal pairs.

Strategy 2: Cross-pool mixing. We merged rays from different algebraic pools, deduplicated by canonical form, and tested whether cross-pool orthogonalities enable smaller KS sets. All 15 pairwise combinations and the full 477-ray union of all six pools were tested with 500 greedy trials each.

Strategy 3: Larger alphabets. We expanded coordinate alphabets beyond those generating the known pools: integer alphabets $\{0, \pm 1, \dots, \pm k\}$ for $k = 3, 4, 5$; extended Peres alphabets adding $\pm 2, \pm 2\sqrt{2}, \pm 3$; Eisenstein pools with increased coefficient bounds and norm cutoffs; extended complex field alphabets for $\mathbb{Z}[\sqrt{-2}]$ and Heegner-7; mixed-field alphabets combining elements from different rings; and completion-expanded pools.

Strategy 4: Numerical optimization. We applied four non-algebraic search methods: (a) simulated annealing (SA) placing $n = 28$ –30 rays in $\mathbb{R}^3$ and $\mathbb{C}^3$, optimizing for orthogonal pairs and triads (200k steps each); (b) perturbation from CK-31, removing 1–3 rays and replacing with random rays (1000 attempts per target); (c) orthogonal completion from random seeds, iteratively adding cross products (500 attempts); (d) soft-tolerance SA starting with tolerance 0.1 and tightening to 10^{-6} over 300k steps.

Strategy 5: CK-31 criticality. We tested whether CK-31 can be reduced by removing k vectors for $k = 1, 2, 3$, exhaustively checking all $\binom{31}{k}$ subsets.

Strategy 6: Triad density analysis. For each pool, we sampled random subsets of varying size and determined the threshold cardinality at which KS-uncolorable

TABLE I. Minimum KS subset size found by each strategy. “Exact” combines MUS, greedy (2000 trials), and swap search. “Cross” is the minimum found when that pool is included in any pairwise or full cross-pool merge (Strategy 2).

Pool	Rays	Triads	Exact	Cross	Larger
Integer	49	26	31	31	31
Eisenstein	57	22	33	33	33
Peres	49	16	33	33	33
$\mathbb{Z}[\sqrt{-2}]$	49	16	33	33	33
Heegner-7	145	42	43	43	43
Golden ratio	205	166	51	51	51

subsets first appear. This characterizes the “rarity” of KS sets within their own pools.

RESULTS

Exact minimization (Strategy 1). All three methods converge to the same minimum for each pool (Table I). No method finds a set below the known minimum. The MUS extraction for the integer pool returns all 49 rays but greedy-minimizes to 31; for $\mathbb{Z}[\sqrt{-2}]$, the MUS directly returns 33 rays.

Cross-pool mixing (Strategy 2). Despite significant cross-pool orthogonalities (e.g., Integer + Golden shares 78 cross-triads), every minimized set draws rays from a single pool only. The composition is always: any combination including Integer $\rightarrow$ 31 rays (all Integer); any combination without Integer $\rightarrow$ 33 rays (all from one pool). The 477-ray union of all six pools minimizes to 31 Integer rays. Cross-pool orthogonalities are never exploited.

Larger alphabets (Strategy 3). We tested 30 expanded alphabets (Table II). Every extension that includes ± 2 converges to 31 using integer rays. Greedy minimization is heuristic, so for large pools (e.g., $\{0, \pm 1, \dots, \pm 4\}$ with 289 rays reaching only 36 in 500 trials) the true minimum may be lower; however, every pool containing the $\{0, \pm 1, \pm 2\}$ subset found 31 when given sufficient trials.

The norm-2 boundary. The alphabets $\{0, \pm 1, \pm 3\}$ and $\{0, \pm 1, \pm 4\}$ —which lack ± 2 —are *not KS-uncolorable at all*, despite generating 49 rays with 10 triads each. This confirms that the cancellation identity $1 + 1 = 2$, which enables exact orthogonalities such as $(1, 1, 0) \cdot (1, -1, 0) = 0$, is essential for KS-uncolorability from integer-like alphabets. More broadly, it supports the norm-2 boundary identified in [4]: every known KS construction relies on a generator whose norm equals 2.

Numerical optimization (Strategy 4). All four methods fail completely. SA from scratch in $\mathbb{R}^3$ produces at most 5–6 orthogonal pairs and *zero* triads after 2×10^5 steps; in $\mathbb{C}^3$, it finds zero pairs. Soft-tolerance SA finds up to 227 near-orthogonal pairs at tolerance 0.1,

TABLE II. Selected expanded alphabets (Strategy 3). “KS” indicates whether the pool is KS-uncolorable; “Min” is the best greedy result in 300–500 trials.

Alphabet	Rays	Min	Note
$\{0, \pm 1, \pm 2\}$	49	31	Baseline (CK-31)
$\{0, \pm 1, \pm 2, \pm 3\}$	145	31	Absorbs to CK-31
$\{0, \pm 1, \pm 2, \pm 3, \pm 4\}$	289	36	Heuristic limit
$\{0, \pm 1, \pm 3\}$	49	—	Not KS
$\{0, \pm 1, \pm 4\}$	49	—	Not KS
Eisenstein $c=3$, $n \leq 6$	345	31	Absorbs to CK-31
Peres $+\pm 2$	109	31	Absorbs to CK-31
Heegner-7 $+\pm 2$	253	31	Absorbs to CK-31
Golden $+\pm 2$	637	33	No absorption
$\{0, \pm 1, \pm 2, \pm i\}$	127	31	Mixed-field

but all vanish as tolerance tightens—zero exact pairs survive. CK-31 perturbation (removing 1–3 rays and adding random replacements) never restores uncolorability: the best replacement achieves only 16 triads versus the 17 required. Orthogonal completion from random seeds builds sets of 60 rays but achieves at most 1 triad. The numerical landscape has no gradient signal: near-orthogonal configurations do not lead to exact ones.

We also tested whether CK-31 can be compressed by *merging* non-orthogonal ray pairs (identifying two vertices) while preserving KS-uncolorability, then checking $\mathbb{R}^3$ realizability using the Z3 SMT solver [6] with non-linear real arithmetic. Of the $\binom{31}{2} - 67 = 398$ non-orthogonal pairs, exactly 15 merges preserve abstract KS-uncolorability, producing 30-vertex graphs with 17–19 triads and 66–67 pairs. However, Z3 could not resolve realizability for any of the 15 within 60 s per instance (non-linear real arithmetic over 90 variables), returning “unknown.” For problems of this type, Z3 typically either finds a solution quickly or does not converge; extending the timeout to 600 s produced no change. This mirrors the computational wall encountered by Li, Bright, and Ganesh [3]: realizability of KS candidate graphs in $\mathbb{R}^3$ is the fundamental bottleneck.

CK-31 criticality (Strategy 5). CK-31 is 1-critical (all 31 single-ray removals tested exhaustively), 2-critical (all 465 pairs), and 3-critical (all 4495 triples)—4991 subsets in total, each verified colorable by SAT. CK-31 cannot be reduced to 30, 29, or 28 vectors by deletion.

Triad density threshold (Strategy 6). In 500 random samples per size, no KS-uncolorable subset of the integer pool appears below $n = 42$ out of 49 rays. For the Eisenstein pool, the threshold is $n = 48$ out of 57. For Heegner-7, no KS subsets are found at any sampled size up to $n = 75$ out of 145. CK-31 achieves the highest pair density among all constructions (15.3% of all $\binom{31}{2}$ pairs are orthogonal). These results show that triad count alone does not determine uncolorability—the topological interlocking of constraints matters.

DISCUSSION

Our results establish three structural facts about KS sets in dimension 3.

First, *the known algebraic pools are self-contained*. Cross-pool mixing never produces a smaller KS set than what exists within a single pool. This suggests that the orthogonality structure of each number ring is optimized for its own KS constructions.

Second, *CK-31 is uniquely minimal* among all tested algebraic sources. No other ring produces a KS set with fewer than 33 vectors, and all paths to 31 lead to integer coordinates. Combined with 3-criticality, this means CK-31 sits at a sharp minimum: it cannot be reduced by deletion, and no known algebraic extension can improve upon it.

Third, *density arguments are insufficient for a proof of optimality*. KS sets are extraordinarily rare even within pools that contain them. The uncolorability depends on the precise topological interlocking of basis constraints—a property that cannot be captured by counting triads or pairs alone. This suggests that closing the 24–31 gap will require realizability-based arguments (extending the Li–Bright–Ganesh framework) rather than combinatorial density bounds.

The restriction to $\mathbb{R}^3$ versus $\mathbb{C}^3$ does not obviously help: working in $\mathbb{C}^3$ provides more orthogonality freedom per ray, but our complex pools (Eisenstein, $\mathbb{Z}[\sqrt{-2}]$, Heegner-7) all minimize to ≥ 33 . The Eisenstein threshold ($n = 48$ out of 57) is higher than the integer threshold ($n = 42$ out of 49), suggesting that the extra complex de-

grees of freedom make it *harder*, not easier, to achieve the tight constraint interlocking needed for small KS sets.

The vertex-merging experiment reveals the nature of the obstacle: abstract KS-uncolorable graphs on 30 vertices exist (15 were found from CK-31 alone), but determining their $\mathbb{R}^3$ realizability exceeds current SMT capabilities. This confirms that the 24–31 gap is fundamentally a realizability problem, not a combinatorial one.

We do not prove that 31 is optimal. However, within the space of algebraically generated ray pools—which encompasses every known KS construction in dimension 3—we have exhausted the search. A sub-31 KS set, if it exists, would require a fundamentally different construction method.

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